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Inception, ResNet, MobileNet

Inception, ResNet, and MobileNet are the convolutional neural networks commonly used for an image classification task. Why such many kinds of networks are needed? The problem behind the development of the networks and the solutions proposed will be explained.

Motivations

Inception is created to serve the purpose of reducing the computational burden of deep neural nets while obtaining state-of-art performance. As the network goes deeper, the computational efficiency will also decrease, therefore the authors of Inception were interested in finding a solution to scale up neural nets without increasing computational cost.

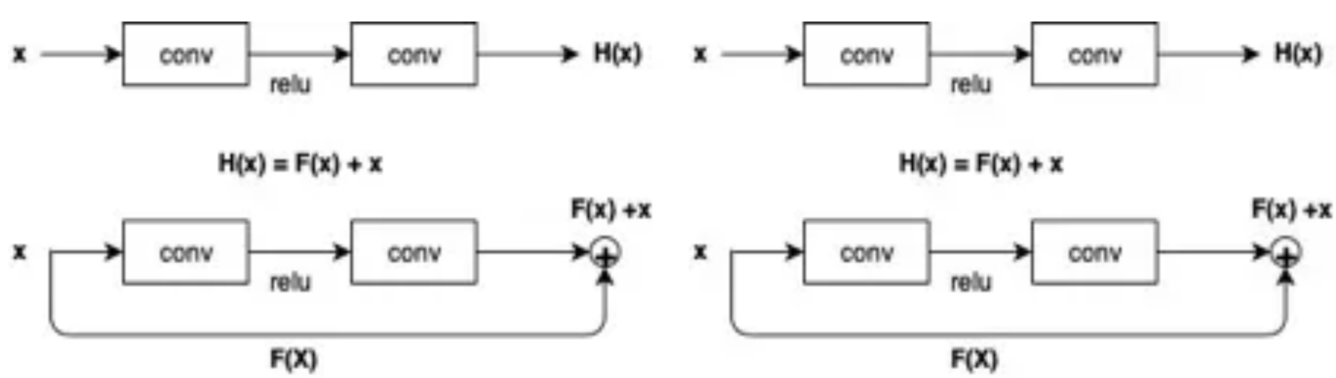
While Inception focuses on computational cost, ResNet focuses on computational accuracy. Intuitively, deeper networks should not perform worse than the shallower networks, but in practice, the deeper networks performed worse than the shallower networks, caused not by overfitting, but by an optimization problem. Shortly, the deeper the network, the harder the network to be optimized.

The trend in the computer vision is to make deeper and more complicated network to achieve higher accuracy. However, deeper networks come with the tradeoff of size and speed. In real applications such as an autonomous vehicle or robotic visions, the object detection task must be able to be done on the computationally limited platform. MobileNet is developed to solve this problem, which is a network for embedded vision applications and mobile devices.

Solutions

The Inception module computes multiple different transformations over the same input map in parallel, connecting the results into a single output. For each layer, it does a 5x5 convolution, 3x3 convolution, and max pooling, each carries different information, which of course is computationally costly. Therefore the authors of Inception decided to overcome this problem by introducing the dimension reductions. What is meant by dimension reduction is by using 1x1 convolution before going to the bottlenecks 3x3 and 5x5 convolutions. Therefore it has the compressed version of the spatial information.

ResNet proposed a solution of using the network layers to fit a residual mapping instead of directly trying to fit a desired underlying mapping.



Top highlight

In other words, as seen in figure above, instead of trying to learn H(x) directly, the network is trying to learn the F(x) + x. This helps the problem of vanishing gradients, in which the gradient signals from the error function decreased exponentially as they are backpropagated. By the time the error signals reached the earlier layer, they were so small and become insignificant. In ResNet, the gradient signal could travel back to early layers via this "shortcut" method, therefore many layers of the network could be created without having accuracy trade-off.

The idea behind MobileNet is to use depthwise separable convolutions to build lighter deep neural networks. In regular convolutional layer, the convolution kernel or filter is applied to all of the channels of the input image, by doing weighted sum of the input pixels with the filter and then slides to the next input pixels across the images. MobileNet uses this regular convolution only in the first layer. The next layers are the depthwise separable convolutions which are the combination of the depthwise and pointwise convolution. The depthwise convolution does the convolution on each channel separately. If the image has three channels, therefore, the output image also has three channels. This depthwise convolution is used to filter the input channels. The next step is the pointwise convolution, which is similar to the regular convolution but with a 1x1 filter. The purpose of pointwise convolution is to merge the output channels of the depthwise convolution to create new features. By doing so, the computational work needed to be done is less than the regular convolutional networks.

Sources:

[1] C. Szegedy *et al.*, "Going deeper with convolutions," *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, vol. 07–12-June, pp. 1–9, 2015.

[2] K. He, "Deep Residual Learning for Image Recognition."

[3] A. G. Howard *et al.*, "MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications," 2017.